

Green Multigene Genetic Programming: Bi-Objective Identification of Polynomial NARX Models under an Arithmetic-Cost Objective

Notas

- **O que mudou no MGGP:** o Green usa seleção de Pareto (NSGA-II), considerando erro e custo juntos. O modo MO-lex prioriza o erro e usa o custo para desempatar.
- **Correção comum:** no estudo novo, um avaliador externo corrige os atrasos de entrada e saída para todos os métodos. Mantém mínimos quadrados e os operadores; simulações que divergem são falhas, não zeros. O pacote original não foi editado.
- **Complexidade:** contamos as operações do polinômio sem termos repetidos: $C_\rho = a + \rho b$, com a adições e b multiplicações, incluindo os coeficientes. Isto é custo aritmético, não energia medida.
- **Dois pesos:** $\rho = 1$ conta todas as operações igualmente; $\rho_K = 729/64$ vem do modelo teórico em bits de Karatsuba. Os dois foram fixados antes das novas buscas e continuam na análise.
- **Medições:** tempos de execução permitem estimar $t_\times/t_+$; powermetrics mede potência para calcular energia; CodeCarbon fornece apenas a intensidade de carbono para calcular CO₂e. Energia e CodeCarbon não definem ρ .
- **Comparações novas:** além do protocolo original, todos os métodos receberam 4.510 avaliações por corrida. Também comparamos objetivos de número de termos e de nós, sem mudar a regra de escolha do modelo.
- **Estado deste documento:** as secções de resultados e energia abaixo ainda são históricas. O estudo corrigido tem 1.440 corridas e relatórios separados em **Resultados/corrected_v1/**; a energia dos novos modelos ainda não foi medida. Os números antigos não podem ser atribuídos aos modelos corrigidos.

1 Green MGGP: Formulation and Theory

This section formalises the model class, the multigene representation and the two objectives, establishes the properties on which Green MGGP relies, and characterises both the lexicographic scheme of the package and the proposed Pareto-based scheme. The main symbols are collected in Table 1, and Fig. 1 gives an overview of the method, from the data to the execution of the selected model, marking the components to which each theoretical result applies.

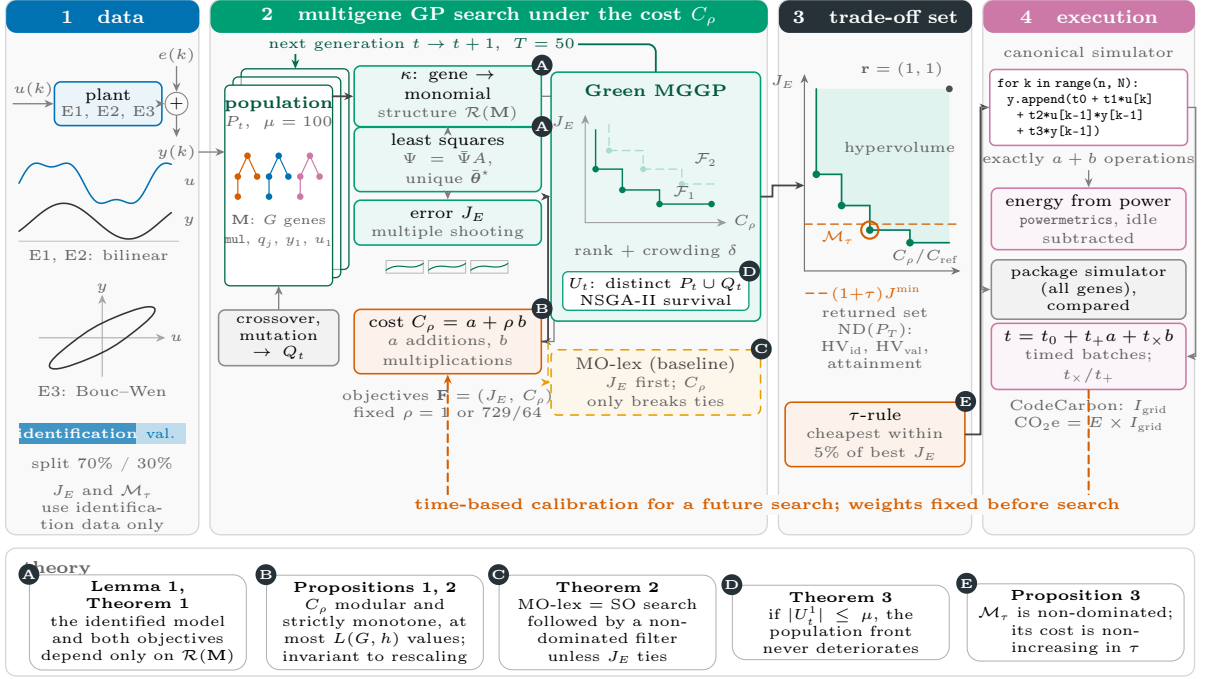


Figure 1: Overview of the methodology. (1) Identification and validation data of the three examples. (2) In Green MGGP, each individual of the population is mapped to its executed structure, estimated by least squares and evaluated by the multiple-shooting error J_E and the arithmetic cost C_ρ ; Pareto ranking with crowding and the removal of structural duplicates select the next population, whereas the MO-lex baseline compares the two objectives lexicographically. (3) The returned non-dominated set is assessed by its hypervolume and reduced to one model by the τ -rule. (4) The selected model is executed as a canonical polynomial, and execution time checks the relation between arithmetic cost and runtime; energy is measured separately. The dashed path denotes time-based calibration for a subsequent study, with its weight fixed before searching; it does not retune the reported searches. Letters A–E mark the components to which the theoretical results apply.

1.1 Monomials and polynomial NARX structures

Polynomial NARX models express the output through monomials of delayed inputs and outputs [1]. The following definitions specify the representation used here. Let $\mathbb{N} = \{0, 1, \dots\}$ and let the input and output be sequences $u, y : \mathbb{Z} \rightarrow \mathbb{R}$. A *factor* $f = (v, \tau) \in \mathbb{F} = \{y, u\} \times \mathbb{N}$ represents the delayed signal $k \mapsto v(k - \tau)$.

Definition 1 (Monomials). A monomial is a finitely supported map $m : \mathbb{F} \rightarrow \mathbb{N}$; the set of monomials $\mathfrak{M}$ is the free commutative monoid generated by $\mathbb{F}$ under pointwise addition, with neutral element 0. Its degree is $|m| = \sum_{f \in \mathbb{F}} m(f)$, its maximum delay is $n(m) = \max\{\tau : m(v, \tau) > 0\}$, and its regressor is

$$\varphi_m(k) = \prod_{(v, \tau) \in \mathbb{F}} v(k - \tau)^{m(v, \tau)}. \quad (1)$$

For $j \in \mathbb{N}$, the delay operator $\sigma_j : \mathfrak{M} \rightarrow \mathfrak{M}$ is $(\sigma_j m)(v, \tau) = m(v, \tau - j)$ if $\tau \geq j$ and 0 otherwise.

Directly from the definition, $\varphi_{m+m'} = \varphi_m \varphi_{m'}$, $\varphi_{\sigma_j m}(k) = \varphi_m(k - j)$, $\sigma_i \sigma_j = \sigma_{i+j}$, $\sigma_j(m + m') = \sigma_j m + \sigma_j m'$ and $|\sigma_j m| = |m|$.

Definition 2 (Structure and model). A polynomial NARX structure is a finite non-empty set $\mathcal{R} \subset \mathfrak{M} \setminus \{0\}$, with maximum delay $n(\mathcal{R}) = \max_{m \in \mathcal{R}} n(m)$. For coefficients $\bar{\theta} = (\theta_0, (\theta_m)_{m \in \mathcal{R}})$,

Table 1: Main symbols.

Symbol	Meaning
$\mathbb{F} = \{y, u\} \times \mathbb{N}$	factors; (v, τ) denotes the delayed signal $k \mapsto v(k - \tau)$
$\mathfrak{M}, m $	monomials (finitely supported $m : \mathbb{F} \rightarrow \mathbb{N}$) and degree $ m = \sum_f m(f)$
φ_m, σ_j	regressor of m ; delay operator on monomials
$\mathcal{T}_h, \mathbf{M} = (g_1, \dots, g_G)$	genes of height at most h ; individual with G genes
$\kappa, \mathcal{R}(\mathbf{M})$	monomial semantics of a gene; executed structure of $\mathbf{M}$
$J_E, C_\rho, \mathbf{F} = (J_E, C_\rho)$	error objective, arithmetic cost, objective vector
$a, b, D = 2^h$	numbers of additions and multiplications, maximum degree
ρ, ρ_K	multiplication weight; bit-level (Karatsuba) weight $B^{\alpha-1}$
$\text{ND}(P), \text{HV}(P)$	non-dominated subset and hypervolume of a finite set P
μ, P_t, U_t	population size, population at generation t , de-duplicated union

the model is

$$\hat{y}(k) = \theta_0 + \sum_{m \in \mathcal{R}} \theta_m \varphi_m(k), \quad (2)$$

which has $p = |\mathcal{R}| + 1$ terms.

1.2 Multigene representation and its semantics

Definition 3 (Genes and individuals). Let $\mathcal{T}$ be the set of finite trees generated by the grammar

$$g ::= y_1 \mid u_1 \mid \text{mul}(g, g) \mid q_j(g), \quad j \in \{1, \dots, n_q\},$$

and let $\mathcal{T}_h \subset \mathcal{T}$ contain the trees of height at most h , leaves having height 0. An individual is a G -tuple $\mathbf{M} = (g_1, \dots, g_G) \in \mathcal{T}_h^G$.

Multigene symbolic regression combines tree outputs through coefficients estimated by least squares [2]. Here we use the representation of the `mggp` package by Ávila dos Santos [3] with the multiplication primitive, where y_1 denotes $y(k - 1)$, u_1 denotes $u(k)$, `mul` is the pointwise product and q_j delays its argument by j samples. Every gene therefore denotes a monomial, as formalised next.

Definition 4 (Monomial semantics). The map $\kappa : \mathcal{T} \rightarrow \mathfrak{M}$ is defined recursively by $\kappa(y_1) = \mathbb{K}_{(y,1)}$, $\kappa(u_1) = \mathbb{K}_{(u,0)}$, $\kappa(\text{mul}(a, b)) = \kappa(a) + \kappa(b)$ and $\kappa(q_j(a)) = \sigma_j \kappa(a)$, where $\mathbb{K}_f$ is the indicator of the factor f . The *executed structure* of an individual is

$$\mathcal{R}(\mathbf{M}) = \{\kappa(g_1), \dots, \kappa(g_G)\}, \quad (3)$$

and two individuals are *structurally equivalent*, $\mathbf{M} \sim \mathbf{M}'$, if $\mathcal{R}(\mathbf{M}) = \mathcal{R}(\mathbf{M}')$.

Lemma 1 (Soundness and invariances of κ). *For every gene $g \in \mathcal{T}$:*

- (a) *the signal computed for g is the regressor $\varphi_{\kappa(g)}$;*
- (b) *κ is invariant under commutation and reassociation of `mul`, under composition of delays $q_i(q_j(a)) \mapsto q_{i+j}(a)$ and under distribution $q_j(\text{mul}(a, b)) \mapsto \text{mul}(q_j(a), q_j(b))$;*
- (c) *$|\kappa(g)|$ equals the number of leaves of g , so $1 \leq |\kappa(g)| \leq 2^h$ for $g \in \mathcal{T}_h$.*

Proof. (a) By structural induction. For the leaves, $\varphi_{\mathbb{K}_{(y,1)}}(k) = y(k - 1)$ and $\varphi_{\mathbb{K}_{(u,0)}}(k) = u(k)$. If the claim holds for a and b , the product computed for `mul`(a, b) is $\varphi_{\kappa(a)} \varphi_{\kappa(b)} = \varphi_{\kappa(a) + \kappa(b)}$, and the signal computed for $q_j(a)$ is $k \mapsto \varphi_{\kappa(a)}(k - j) = \varphi_{\sigma_j \kappa(a)}(k)$. (b) The four transformations are mapped by κ to the identities $m + m' = m' + m$, $(m + m') + m'' = m + (m' + m'')$, $\sigma_i \sigma_j = \sigma_{i+j}$ and $\sigma_j(m + m') = \sigma_j m + \sigma_j m'$ in $\mathfrak{M}$. (c) Each leaf contributes one unit of degree, `mul` adds degrees and σ_j preserves them. A tree of height h whose only binary node is `mul` has at most 2^h leaves. $\square$

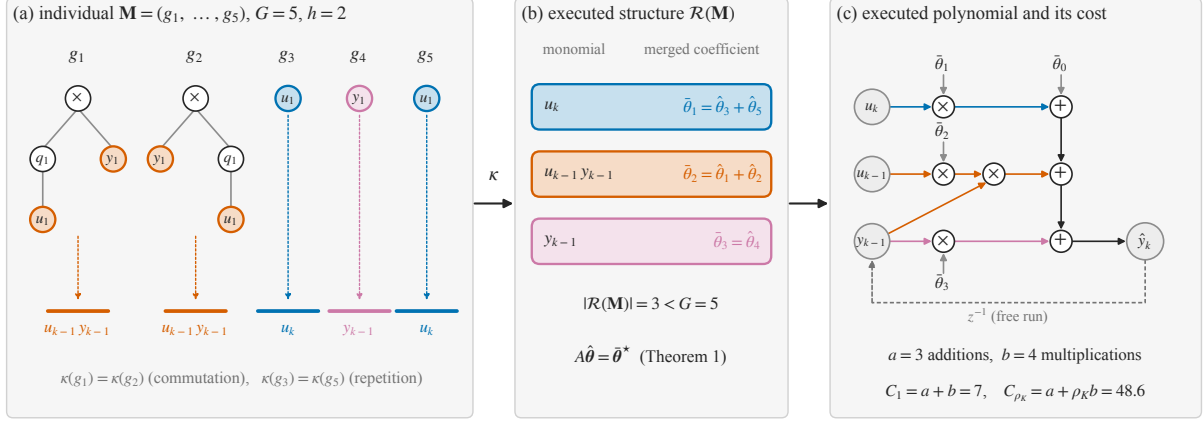


Figure 2: From an individual to its arithmetic cost, for a model returned by Green MGGP on E1 ($\rho = 1$, seed 13). (a) The five gene trees, with leaves coloured by the monomial that the gene denotes: g_1 and g_2 differ only by commutation, and g_3 and g_5 coincide. (b) The executed structure $\mathcal{R}(\mathbf{M})$ has three monomials, and the coefficients of genes with the same monomial are merged (Theorem 1). (c) The executed polynomial, simulated in free run, needs $a = 3$ additions and $b = 4$ multiplications per sample, so that $C_1 = 7$ and $C_{\rho_K} = 48.6$ in (5). This illustrative model omits the true bilinear regressor and is not the generating structure.

Lemma 1(b) shows that many syntactically different individuals execute the same polynomial; for instance $\text{mul}(q_1(y_1), u_1)$ and $\text{mul}(u_1, q_1(y_1))$ both denote $y(k-2)u(k)$. The map $\mathbf{M} \mapsto \mathcal{R}(\mathbf{M})$ removes this redundancy, and $|\mathcal{R}(\mathbf{M})| \leq G$, with strict inequality when genes coincide. Fig. 2 illustrates these notions on an individual returned in one of the runs of Section 3: its five genes execute only three monomials, so the polynomial that is actually evaluated, and whose operations define the cost introduced below, is smaller than the representation suggests.

1.3 Parameter estimation and structural invariance

Let N identification samples be given and let $\mathcal{K} = \{n+1, \dots, N\}$, where n is the maximum delay of the individual. The regression matrix $\Psi(\mathbf{M}) \in \mathbb{R}^{|\mathcal{K}| \times (G+1)}$ has a column of ones and the columns $(\varphi_{\kappa(g_i)}(k))_{k \in \mathcal{K}}$, and $\bar{\Psi}(\mathcal{R}) \in \mathbb{R}^{|\mathcal{K}| \times (|\mathcal{R}|+1)}$ is the corresponding matrix with one column per element of $\mathcal{R}$.

Assumption 1 (Identifiability). $\bar{\Psi}(\mathcal{R})$ has full column rank.

Theorem 1 (Structural invariance of the identified model). *Let $\mathbf{M}$ be an individual with $\mathcal{R} = \mathcal{R}(\mathbf{M})$ satisfying Assumption 1, and let $A \in \{0, 1\}^{(|\mathcal{R}|+1) \times (G+1)}$ be the aggregation matrix with $A_{00} = 1$ and $A_{mi} = 1$ if and only if $\kappa(g_i) = m$. Then:*

- (i) $\Psi(\mathbf{M}) = \bar{\Psi}(\mathcal{R})A$;
- (ii) every minimiser $\hat{\boldsymbol{\theta}}$ of $\|\mathbf{y} - \Psi(\mathbf{M})\boldsymbol{\theta}\|_2$ satisfies $A\hat{\boldsymbol{\theta}} = \bar{\boldsymbol{\theta}}^*$, the unique minimiser of $\|\mathbf{y} - \bar{\Psi}(\mathcal{R})\bar{\boldsymbol{\theta}}\|_2$;
- (iii) the identified model of $\mathbf{M}$ coincides with the model (2) of $\mathcal{R}$ with coefficients $\bar{\boldsymbol{\theta}}^*$. Consequently its one-step-ahead, multiple-shooting and free-run predictions, and every objective computed from them, depend on $\mathbf{M}$ only through $\mathcal{R}(\mathbf{M})$.

Proof. (i) Column $i \geq 1$ of $\Psi(\mathbf{M})$ is the column of $\bar{\Psi}(\mathcal{R})$ indexed by $\kappa(g_i)$, that is, $\Psi e_i = \bar{\Psi} e_{\kappa(g_i)} = \bar{\Psi} A e_i$; the constant columns agree because $A_{00} = 1$. (ii) Every row of A contains a one, because every $m \in \mathcal{R}$ equals $\kappa(g_i)$ for some i , so A is surjective. Hence $\{\Psi\boldsymbol{\theta}\} = \{\bar{\Psi}A\boldsymbol{\theta}\} = \{\bar{\Psi}\bar{\boldsymbol{\theta}}\}$, and the two least-squares problems have the same optimal value. If $\hat{\boldsymbol{\theta}}$ is a minimiser of the

first, $A\hat{\theta}$ attains the optimal value of the second and is therefore a minimiser of it. Under Assumption 1 the second objective is strictly convex, so its minimiser is unique. (iii) For every argument, $\hat{\theta}_0 + \sum_{i=1}^G \hat{\theta}_i \varphi_{\kappa(g_i)} = \hat{\theta}_0 + \sum_{m \in \mathcal{R}} (\sum_{i: \kappa(g_i)=m} \hat{\theta}_i) \varphi_m = \theta_0^* + \sum_{m \in \mathcal{R}} \theta_m^* \varphi_m$. All three predictors evaluate this function recursively from the data and from initial conditions of length $n(\mathcal{R})$, which is itself a function of $\mathcal{R}$. $\square$

Corollary 1. *Every objective that is a function of the identified model and of $\mathcal{R}(\mathbf{M})$ is constant on the equivalence classes of $\sim$. Removing all but one member of each class from a set of individuals therefore removes no objective vector.*

1.4 Error objective

Multiple shooting limits simulation length by dividing the data into shorter segments [4]. Here the package resets each segment to measured outputs; this windowed criterion does not include the optimised segment initial states and continuity constraints of the formulation in [4].

Definition 5 (Multiple-shooting error). Let $n = n(\mathcal{R}(\mathbf{M}))$, $K \in \mathbb{N}$ and $W = \lfloor N/(n + K) \rfloor$. The identification samples are partitioned into W disjoint windows of length $n + K$. In window w , the identified model is initialised with the n measured outputs and simulated in free run for K steps, giving $\hat{y}_w(k)$. The error objective is

$$J_E(\mathbf{M}) = \frac{1}{\sigma_y} \left(\frac{1}{W(n + K)} \sum_{w=1}^W \sum_{k=1}^{n+K} (y_w(k) - \hat{y}_w(k))^2 \right)^{1/2}, \quad (4)$$

where σ_y is the standard deviation of the identification output, and $J_E(\mathbf{M}) = \infty$ if the estimation or the simulation fails.

The first n residuals of each window vanish by construction. By Theorem 1, J_E is a function of $\mathcal{R}(\mathbf{M})$. With $K = 20$ it penalises the accumulation of error over a finite horizon at a fraction of the cost of a full free-run simulation; the normalisation by σ_y does not affect the search.

The definitions and structural-invariance results above describe the intended polynomial predictors. The unmodified package has a circular-buffer error in multiple shooting and free run: it retains only the maximum backshift in past outputs, so an output factor at that boundary can be evaluated as $y(k - 1)$ instead of its intended delay. Least-squares estimation uses the intended delays. For affected individuals, the numerical error used by the search therefore differs from (4) evaluated with the intended polynomial. All configurations retain this implementation, and the stored objectives are reported without correction. Input-only individuals without a backshift can be evaluated; the failure for no backshift concerns individuals containing output factors. These implementation restrictions qualify the empirical application of the ideal predictor definitions.

1.5 Arithmetic-cost objective

Definition 6 (Arithmetic cost). Evaluating (2) in direct form requires $a(\mathcal{R}) = |\mathcal{R}|$ additions and $b(\mathcal{R}) = \sum_{m \in \mathcal{R}} |m|$ multiplications, one for each coefficient and $|m| - 1$ between the factors of each regressor. Let $\rho > 0$ be the cost of a multiplication relative to that of an addition on the platform that executes the model. In addition-equivalents, the cost of a regressor and of an individual are

$$c_\rho(m) = 1 + \rho |m|, \quad C_\rho(\mathbf{M}) = \sum_{m \in \mathcal{R}(\mathbf{M})} c_\rho(m) = a(\mathcal{R}) + \rho b(\mathcal{R}). \quad (5)$$

Two weights are used. The bit-level weight of the Green-Box framework [5] charges B bit operations for an addition of B -bit operands and B^α , $\alpha = \log_2 3$, for a Karatsuba multiplication [6]; it corresponds to $\rho_K = B^{\alpha-1}$, which for binary64 operands ($B = 64$) equals

$3^6/2^6 = 729/64 \approx 11.39$, and $B C_{\rho_K} = B a + B^\alpha b$ is the Green-Box complexity. The unit weight $\rho = 1$ makes $C_1 = a + b$ the number of arithmetic operations, which describes platforms on which both operations take comparable time, such as processors with floating-point units; Section 4 shows that it is the weight supported by the execution times measured in this study. We write C for C_ρ when the weight is clear.

1.5.1 Establishing the weights and separating the measurements

The two declared scenarios are analysed throughout: $\rho = 1$ is an unweighted operation count and $\rho_K = 729/64$ is a theoretical bit-level weight. Neither is obtained from power measurements. The bit-level model does not describe the instruction-level cost of binary64 multiplication on this laptop. Operation counts are computed from each canonical structure before evaluation, independently of any measured runtime, energy or carbon intensity.

A platform-specific timing calibration instead estimates $\rho_{\text{time}} = t_\times/t_+$ from $t = t_0 + t_+ a + t_\times b$. Here this fit characterises the Python simulator and includes interpreter overhead. Earlier timing measurements provide support for an order-one weight; they do not make $\rho = 1$ an exact fitted value. In the corrected study both declared weights were fixed before the new searches. Subsequent timing checks assess those choices and cannot retrospectively redefine their objectives. A newly calibrated weight would require a separately declared search protocol. Section 4 separates timing, power integration and carbon conversion; its historical measurements are not measurements of the corrected models.

Proposition 1 (Structure of the cost). *Let $\rho > 0$ and let $\mathcal{R}, \mathcal{R}'$ be structures of individuals in $\mathcal{T}_h^G$, with $D = 2^h$:*

- (i) C_ρ is modular: $C_\rho(\mathcal{R} \cup \mathcal{R}') = C_\rho(\mathcal{R}) + C_\rho(\mathcal{R}') - C_\rho(\mathcal{R} \cap \mathcal{R}')$;
- (ii) C_ρ is strictly monotone: $\mathcal{R} \subsetneq \mathcal{R}'$ implies $C_\rho(\mathcal{R}) < C_\rho(\mathcal{R}')$, and $|m| < |m'|$ implies $c_\rho(m) < c_\rho(m')$;
- (iii) $(1 + \rho) a(\mathcal{R}) \leq C_\rho(\mathcal{R}) \leq (1 + \rho D) a(\mathcal{R})$;
- (iv) C_ρ depends on $\mathcal{R}$ only through (a, b) , and the number of distinct values of C_ρ over $\mathcal{T}_h^G$ is at most
$$L(G, h) = \sum_{a=1}^G (a(D-1) + 1) = G + \frac{(D-1)G(G+1)}{2}; \quad (6)$$
- (v) for $\rho = \rho_K$ and $G < 729$, the map $(a, b) \mapsto C_\rho$ is injective, so the cost determines both the number of terms and the total degree; for $\rho = 1$ it is not, and C_1 takes at most $G(D+1) - 1$ values.

Proof. (i) C_ρ is a sum of c_ρ over the elements of a set. (ii) $c_\rho(m) > 0$ for every m , and c_ρ is strictly increasing in $|m|$. (iii) By Lemma 1(c), $1 \leq |m| \leq D$. (iv) The first statement is (5); moreover $a \in \{1, \dots, G\}$ and, by (iii), $a \leq b \leq aD$, which gives at most $a(D-1) + 1$ values of b for each a . (v) $\rho_K = 64^{\log_2 3 - 1} = 3^6/2^6$. If $a + \rho_K b = a' + \rho_K b'$, then $64(a - a') = 729(b' - b)$; since $\gcd(64, 729) = 1$, 729 divides $a - a'$, and $|a - a'| < G < 729$ forces $a = a'$ and then $b = b'$. For $\rho = 1$, $(a, b) = (2, 2)$ and $(1, 3)$ have the same cost, and $C_1 = a + b$ ranges over the integers from 2 to $G(D+1)$. $\square$

Remark 1. The complexity used in the tutorial of the package, ω times the number of `mul` nodes, assigns $c_\omega(m) = \omega(|m| - 1)$ to each gene and sums over genes rather than over $\mathcal{R}(\mathbf{M})$. It violates Proposition 1(ii), because a linear regressor costs nothing, so adding linear terms never increases the cost, and it charges duplicated genes that are executed once.

Proposition 2 (Invariance to monotone rescaling). *Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing. The Pareto order and the lexicographic order induced by $(J_E, \psi \circ C)$ coincide with those induced by (J_E, C) . In particular, any constant factor $\omega > 0$ applied to a cost has no effect on an algorithm whose decisions depend only on these orders, and the Green-Box complexity BC_{ρ_K} induces the same orders as C_{ρ_K} .*

Proof. ψ preserves $\leq$, $<$ and $=$, and both orders are defined from these relations applied componentwise. $\square$

The weight ρ , in contrast, is not a rescaling: it changes the relative cost of structures with different numbers of terms and degrees, and hence the orders. A structure with fewer, higher-degree terms can be cheaper than one with more, linear terms under $\rho = 1$ and more expensive under ρ_K , so the weight must reflect the target platform.

1.6 Bi-objective problem

We use Pareto dominance to compare the objectives and hypervolume to measure the dominated region [7].

Definition 7 (Dominance, Pareto set, hypervolume). For $\mathbf{F} = (J_E, C)$, $\mathbf{M}$ dominates $\mathbf{M}'$, written $\mathbf{M} \prec \mathbf{M}'$, if $\mathbf{F}(\mathbf{M}) \leq \mathbf{F}(\mathbf{M}')$ componentwise and $\mathbf{F}(\mathbf{M}) \neq \mathbf{F}(\mathbf{M}')$. For a finite set P , $\text{ND}(P)$ is its subset of non-dominated elements. The problem is

$$\min_{\mathbf{M} \in \mathcal{T}_h^G} \mathbf{F}(\mathbf{M}) = (J_E(\mathbf{M}), C(\mathbf{M})), \quad (7)$$

whose solution is the set of non-dominated individuals. For a reference point $\mathbf{r}$, the hypervolume of P is the Lebesgue measure $\text{HV}(P) = \lambda(\bigcup_{\mathbf{M} \in P} [\mathbf{F}(\mathbf{M}), \mathbf{r}])$.

If every element of P is weakly dominated by an element of Q , then the region dominated by P is contained in that dominated by Q , so $\text{HV}(P) \leq \text{HV}(Q)$; this monotonicity is used below.

1.7 Selection schemes

Both schemes share one generation template: parent selection by binary tournament, crossover and mutation with the operators of the package, least-squares estimation and evaluation of the modified offspring, and elitist survival. They differ only in the relation used to compare individuals and in the survival rule.

Lexicographic scheme (MO-lex). In the bi-objective mode of the package [3], tournaments and the hall of fame use DEAP fitness comparisons [8, 9], which follow the lexicographic order: $\mathbf{F}(\mathbf{M}) <_{\text{lex}} \mathbf{F}(\mathbf{M}')$ if $J_E(\mathbf{M}) < J_E(\mathbf{M}')$, or $J_E(\mathbf{M}) = J_E(\mathbf{M}')$ and $C(\mathbf{M}) < C(\mathbf{M}')$. The single-objective configuration (SO) compares J_E only. The run returns $\text{ND}(P_T)$ in MO-lex and the best individual in SO, where P_T is the final population.

Theorem 2 (Lexicographic reduction). *Run SO and MO-lex from the same initial population with the same pseudo-random stream. Suppose that, up to generation t , every pair of individuals compared by a tournament or by the hall of fame with $J_E(\mathbf{M}) = J_E(\mathbf{M}')$ also satisfies $C(\mathbf{M}) = C(\mathbf{M}')$. Then both runs produce the same populations $P_0, \dots, P_t$. If the condition holds for all generations, MO-lex returns $\text{ND}(P_T^{\text{SO}})$, the non-dominated subset of the final SO population.*

Proof. By induction on the generation. P_0 is common. Suppose that P_s is common. A generation is a deterministic function of the current population, of the pseudo-random numbers it consumes and of the outcomes of the comparisons it performs; the number and order of the random draws depend only on the population and on these outcomes. For a compared pair with $J_E(\mathbf{M}) \neq$

Algorithm 1 One generation of Green MGGP

Require: population P_t with $|P_t| = \mu$; crossover and mutation rates p_c, p_m

- 1: assign rank and crowding distance (8) to all $\mathbf{M} \in P_t$
 - 2: $Q_t \leftarrow \mu$ copies selected by binary tournament under the crowded order
 - 3: apply package crossover to consecutive pairs of Q_t with probability p_c
 - 4: apply package mutation to each $\mathbf{M} \in Q_t$ with probability p_m
 - 5: estimate $\hat{\theta}$ and evaluate $\mathbf{F}$ for every modified individual
 - 6: $U_t \leftarrow$ one representative per class of $\sim$ among the elements of $P_t \cup Q_t$ with finite $\mathbf{F}$
 - 7: $P_{t+1} \leftarrow$ the first μ elements of U_t under the crowded order (NSGA-II truncation); if $|U_t| < \mu$, complete with the remaining individuals
-

$J_E(\mathbf{M}')$, both relations order the pair as J_E does. For a pair with $J_E(\mathbf{M}) = J_E(\mathbf{M}')$, the hypothesis gives $C(\mathbf{M}) = C(\mathbf{M}')$, so both relations declare a tie, which the implementation resolves by position in the same way. All outcomes coincide, the same random numbers are consumed, the same offspring are produced and evaluated deterministically, and P_{s+1} is common. The adaptation of the mutation rate depends on J_E of the best individual and is therefore also common. The last statement follows from $P_T^{\text{lex}} = P_T^{\text{SO}}$. $\square$

Theorem 2 states that the lexicographic scheme is the single-objective algorithm followed by a non-dominated filter, unless exact ties in J_E between individuals with different costs occur. Since J_E is a continuous function of the data, such ties essentially arise only among individuals whose estimation or simulation fails ($J_E = \infty$); Section 3 verifies the theorem empirically.

Pareto-based scheme (Green MGGP). Green MGGP replaces the comparison relation and the survival rule by those of NSGA-II [10].

Definition 8 (Rank and crowding). For a finite set P , let $\mathcal{F}_1 = \text{ND}(P)$ and $\mathcal{F}_{r+1} = \text{ND}(P \setminus \bigcup_{s \leq r} \mathcal{F}_s)$. The rank of $\mathbf{M} \in \mathcal{F}_r$ is r , and its crowding distance is

$$\delta(\mathbf{M}) = \sum_{i=1}^2 \frac{F_i(\mathbf{M}_i^+) - F_i(\mathbf{M}_i^-)}{F_i^{\max} - F_i^{\min}}, \quad (8)$$

where $\mathbf{M}_i^\pm$ are the neighbours of $\mathbf{M}$ in $\mathcal{F}_r$ sorted by F_i , the extremes and normalisation are taken over $\mathcal{F}_r$, and boundary elements receive $\delta = \infty$. The crowded order prefers lower rank and, for equal ranks, larger δ .

Theorem 3 (Non-deterioration of the population front). *Let $U_t^1 = \text{ND}(U_t)$ in Algorithm 1 and suppose that P_t contains an individual with finite $\mathbf{F}$. If $|U_t^1| \leq \mu$, then:*

- (i) $U_t^1 \subseteq P_{t+1}$;
- (ii) for every $\mathbf{M} \in \text{ND}(P_t)$ there is $\mathbf{M}' \in \text{ND}(P_{t+1})$ with $\mathbf{F}(\mathbf{M}') \leq \mathbf{F}(\mathbf{M})$;
- (iii) $\text{HV}(\text{ND}(P_{t+1})) \geq \text{HV}(\text{ND}(P_t))$ for every reference point.

Moreover, if no two elements of U_t^1 share an objective vector, then $|U_t^1| \leq L(G, h)$ in (6), so $L(G, h) \leq \mu$ is a sufficient condition for (i)–(iii) at every generation.

Proof. (i) NSGA-II truncation admits the fronts of U_t in increasing rank, and $\mathcal{F}_1 = U_t^1$ fits entirely. (ii) Let $\mathbf{M} \in \text{ND}(P_t)$. Since P_t has an element with finite $\mathbf{F}$, so does $\mathbf{M}$. Its class has a representative $\tilde{\mathbf{M}} \in U_t$ with $\mathbf{F}(\tilde{\mathbf{M}}) = \mathbf{F}(\mathbf{M})$ (Corollary 1). Because U_t is finite and $\prec$ is a strict partial order, either $\tilde{\mathbf{M}} \in U_t^1$ or some $\mathbf{M}' \in U_t^1$ dominates $\tilde{\mathbf{M}}$; in both cases there is $\mathbf{M}' \in U_t^1$ with $\mathbf{F}(\mathbf{M}') \leq \mathbf{F}(\mathbf{M})$, and $\mathbf{M}' \in P_{t+1}$ by (i). Every element of P_{t+1} either belongs to U_t , shares

its objective vector with an element of U_t , or has $\mathbf{F} = (\infty, \infty)$; none of them dominates $\mathbf{M}'$, so $\mathbf{M}' \in \text{ND}(P_{t+1})$. (iii) follows from (ii) and the monotonicity of the hypervolume. For the bound, two elements of a non-dominated set with the same cost must have the same error, since otherwise one dominates the other; when no two elements share an objective vector, U_t^1 has at most one element per value of C_ρ , and Proposition 1(iv) bounds the number of these values by $L(G, h)$ for every $\rho > 0$. $\square$

The single-objective and lexicographic schemes enjoy no such guarantee for the cost dimension: their elitism retains the individuals with the lowest J_E , so a cheap non-dominated individual can be lost from one generation to the next.

1.8 Selecting a single model

Definition 9 (τ -rule). For a returned set P , $\tau \geq 0$ and $\eta \geq 0$, let $J^{\min} = \min_{\mathbf{M} \in P} J_E(\mathbf{M})$ and

$$\mathcal{M}_\tau \in \arg \min \{ C(\mathbf{M}) : \mathbf{M} \in P, J_E(\mathbf{M}) \leq (1 + \tau)J^{\min} + \eta \}, \quad (9)$$

with ties broken by the smaller J_E .

Proposition 3 (Properties of the τ -rule). (i) $\mathcal{M}_\tau \in \text{ND}(P)$. (ii) $\tau \mapsto C(\mathcal{M}_\tau)$ is non-increasing; if $P = \text{ND}(P)$, then $\tau \mapsto J_E(\mathcal{M}_\tau)$ is non-decreasing. (iii) $\mathcal{M}_0$ is the cheapest model of minimal error up to η , and for τ large enough $\mathcal{M}_\tau$ is the cheapest model in P .

Proof. (i) Suppose $\mathbf{M} \prec \mathcal{M}_\tau$. Then $J_E(\mathbf{M}) \leq J_E(\mathcal{M}_\tau)$, so $\mathbf{M}$ is feasible, and $C(\mathbf{M}) \leq C(\mathcal{M}_\tau)$. Minimality of the cost forces $C(\mathbf{M}) = C(\mathcal{M}_\tau)$, and then $J_E(\mathbf{M}) < J_E(\mathcal{M}_\tau)$ contradicts the tie-breaking rule. (ii) The feasible set grows with τ , so its minimum cost cannot increase. If P is non-dominated, a smaller cost implies a larger error, and an equal cost implies an equal vector. (iii) is immediate from the definition. $\square$

We use $\tau = 0.05$ and $\eta = 10^{-6}$, fixed before the experiments; $\mathcal{M}_\tau$ is then the cheapest model whose identification error lies within 5% of the best error found, and η only matters on noise-free data, where the best error is at machine precision. For SO, P contains a single model. The rule uses identification data only, and its sensitivity to τ is examined in Section 3.

2 Experimental Setup

The results reported below retain the historical evaluator and caches. The corrected study is stored separately and applies a common external evaluator to every method, with both multiplication weights. Its equal-evaluation arm includes term-count and representation-node ablations. Historical energy measurements are kept separate from these newly selected models.

2.1 Examples

The study uses the regression examples distributed with the `mggp` package [3] (Table 2, Fig. 3); the classification mode of the package returns a single objective and is not considered. Example E1 is the tutorial system

$$y(k) = 0.65 y(k-1) + 0.35 u(k) + 0.10 u(k) y(k-1), \quad (10)$$

excited by $u = \sin t + 0.25 \sin 3t$ over 300 samples, of which the first 70% are used for identification and the remainder for validation. Example E2 adds white Gaussian measurement noise with standard deviation 0.02 to the output of E1. Example E3 uses the package simulator of the Bouc–Wen hysteresis model [11], $y = d_p u - h$ with $\dot{h} = A\dot{u} - \beta|\dot{u}|h - \gamma\dot{u}|h|$, $A = 0.9$, $\beta = \gamma = 0.008$ and $d_p = 1.6$, integrated by fourth-order Runge–Kutta with a step of 1 ms. Identification uses a 1 Hz sinusoid whose amplitude grows linearly from 20 to 50 over 6 s, and validation a 0.8 Hz sinusoid of amplitude 45 over 4 s; both are decimated by 20.

Table 2: Examples and search settings; C_{ref} for $\rho = 1$ (common: population 100, 50 generations, crossover 0.9, initial mutation 0.2; baseline elite retention 10%, multiple-shooting horizon $K = 20$, primitives mul and q^{-j}).

Example	N_{id}	N_{val}	genes	height	j_{max}	C_{ref}
E1 Tutorial (noise-free)	210	90	5	2	2	25
E2 Tutorial (noisy)	210	90	5	2	2	25
E3 Bouc–Wen hysteresis	300	200	8	3	3	72

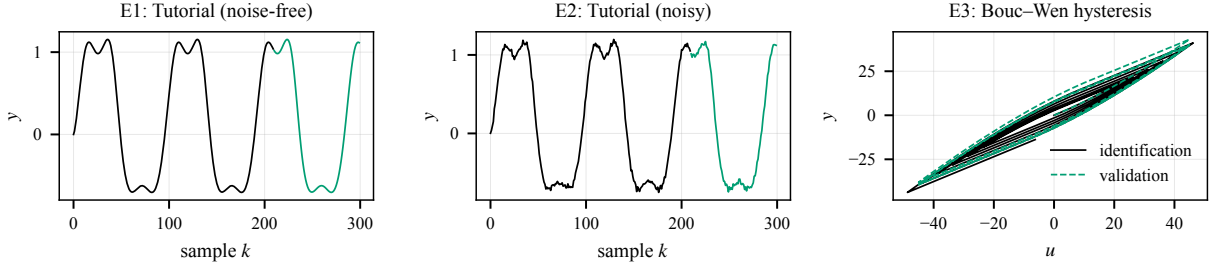


Figure 3: Identification and validation data of the three examples. E3 is shown in the input–output plane, where the loops reveal the hysteresis.

2.2 Compared methods and protocol

Three configurations share the representation, operators, initial rates and population/generation settings of Table 2: SO, the single-objective package minimising J_E , which returns its best model; MO-lex, the bi-objective package mode with C_ρ in (5) as the second objective; and the proposed Green MGGP. Each configuration is run with 30 seeds per example under two multiplication weights: the unit weight $\rho = 1$, which is the primary operation-count setting, supported in scale by the historical execution times (Section 4), and the bit-level Green-Box weight $\rho_K = 11.39$. SO does not use the cost during the search, so its runs are common to both settings. Unless stated otherwise, the results refer to $\rho = 1$. The returned set of each run (one model for SO) is reduced to distinct structures, and each model is refitted on the identification data and simulated in free run on the validation data, initialised with the first n measured outputs.

The 50 recorded generations comprise the initial population and 49 evolutionary steps. With 10% elite retention, SO and MO-lex generate 90 offspring per step, whereas Green generates 100 before union-based survival. Thus equal generation counts do not imply equal numbers of fitness evaluations; the actual number also depends on reuse of valid fitness values. The mutation probability starts at 0.2 and follows the package’s adaptive schedule, increasing by 0.1, up to 0.9, at nine-generation checkpoints when improvement is below 5%. Elite retention of 10% applies to the baselines; Green uses elitist union-based survival.

Each returned set is assessed by its hypervolume [7] in the normalised plane whose axes are C/C_{ref} and NRMSE, with reference point (1,1), where C_{ref} is the cost of G genes of the maximum degree $2^{h_{\text{max}}}$ (Table 2). HV_{id} uses J_E and HV_{val} the free-run validation NRMSE of the same models; points beyond the reference contribute nothing. The model $\mathcal{M}_\tau$ in (9) is characterised by its cost C_τ and validation NRMSE. For E1 and E2 we also record whether $\mathcal{M}_\tau$ contains the three regressors of (10). Green MGGP is compared with each baseline by two-sided Mann–Whitney tests [12] with Holm correction [13] over all comparisons, and effect sizes are reported by the Vargha–Delaney statistic A_{12} [14]. All runs were executed in Python 3.13 on a single CPU core.

Table 3: Results over 30 independent runs with $\rho = 1$: median [interquartile range]. HV: hypervolume of the delivered set (identification or free-run validation error); C_τ , NRMSE_τ : cost and free-run validation NRMSE of the model selected by the τ -rule; Rec_τ , $\text{Rec}_\mathcal{P}$: fraction of runs in which the selected model, or at least one model of the returned set, contains the three true regressors of (10). $^*/^\dagger$: Green differs from SO/MO-lex (Mann–Whitney, Holm-adjusted $p < 0.05$).

Ex.	Method	HV _{id}	HV _{val}	C_τ	NRMSE_τ	Rec_τ	$\text{Rec}_\mathcal{P}$
E1	SO	0.560 [0.490, 0.600]	0.560 [0.490, 0.600]	11 [10, 12.8]	4e-16 [3e-16, 4e-16]	1.00	1.00
	MO-lex	0.750 [0.719, 0.756]	0.752 [0.722, 0.755]	9 [9, 9]	4e-16 [4e-16, 8e-16]	1.00	1.00
	Green	0.895 [0.895, 0.895] *†	0.893 [0.893, 0.893] *†	9 [9, 9] *	1e-15 [8e-16, 2e-15] *†	1.00	1.00
E2	SO	0.507 [0.468, 0.507]	0.504 [0.465, 0.504]	12 [12, 13]	0.031 [0.031, 0.031]	0.00	0.00
	MO-lex	0.733 [0.701, 0.735]	0.729 [0.702, 0.731]	10 [10, 12]	0.031 [0.031, 0.031]	0.00	0.13
	Green	0.876 [0.875, 0.876] *†	0.871 [0.871, 0.871] *†	10 [10, 10] *†	0.031 [0.031, 0.031]	0.00	1.00
E3	SO	0.698 [0.688, 0.710]	0.661 [0.638, 0.675]	20.5 [20, 22]	0.067 [0.058, 0.080]	–	–
	MO-lex	0.792 [0.772, 0.809]	0.757 [0.739, 0.783]	20.5 [16.5, 22]	0.066 [0.057, 0.079]	–	–
	Green	0.934 [0.920, 0.939] *†	0.897 [0.889, 0.912] *†	14 [8.5, 16] *†	0.078 [0.063, 0.081]	–	–

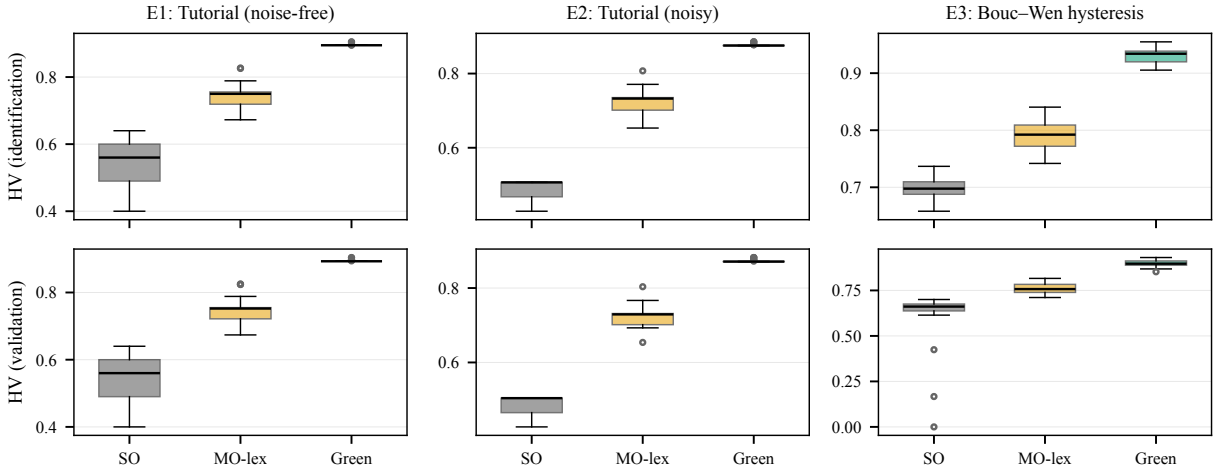


Figure 4: Hypervolume of the returned sets over 30 runs with $\rho = 1$, computed with the identification error (top) and the free-run validation error (bottom) of the same models.

3 Results and Discussion

This section reports the historical cached runs with the unit weight $\rho = 1$; Section 3.5 compares them with the runs under the bit-level weight ρ_K . Both costs are computed from model structures; neither uses measured energy. The corrected-study comparisons remain in their separate reports and are not combined with these historical numbers.

3.1 Quality of the trade-off sets

Table 3 summarises the 30 runs per configuration. On all three examples, Green MGGP returns sets with higher hypervolume than both baselines, in identification (0.895, 0.876 and 0.934 against 0.750, 0.733 and 0.792 for MO-lex) and in free-run validation (0.893, 0.871 and 0.897 against 0.752, 0.729 and 0.757). Every Green run exceeds every baseline run on these indicators with the declared reference (1, 1) ($A_{12} = 1.00$, Holm-adjusted $p < 10^{-8}$ for all twelve comparisons), and the Green interquartile ranges are also the narrowest (Fig. 4). The validation hypervolumes closely follow the identification ones, so the additional low-cost models found by Green MGGP are not artefacts of the identification criterion.

Table 4: Post-hoc sensitivity of validation hypervolume to its error reference ($\rho = 1$). A_{12} compares Green with MO-lex; the cost reference is unchanged.

Example	Error reference	Green median	A_{12}
E1	1	0.893	1.00
E1	0.2	0.157	1.00
E1	0.1	0.071	1.00
E2	1	0.871	1.00
E2	0.2	0.135	1.00
E2	0.1	0.051	0.95
E3	1	0.897	1.00
E3	0.2	0.129	0.81
E3	0.1	0.034	0.52

The SO comparison concerns its delivered single model, whereas MO-lex and Green return sets. As a diagnostic, the median identification hypervolume of the non-dominated final SO population is 0.753, 0.735 and 0.789 on E1–E3, close to MO-lex. This separates the effect of final filtering from the search itself without changing the declared SO output.

The complete separation of hypervolumes is conditional on the declared reference. Reference-point specification affects the relative contributions of different regions of a trade-off set [15]. Table 4 recomputes validation hypervolume from the stored models with tighter error references, as a post-hoc sensitivity analysis. On E3, the Green–MO-lex effect approaches equality at an error reference of 0.1. Thus the primary indicator rewards the broad accuracy–cost trade-off, and does not establish superiority within every restricted high-accuracy region.

The hypervolume condenses each set into one number; Fig. 5 shows where the differences lie. For each run r and cost level c , let $A_r(c)$ be the smallest error among the returned models with $C \leq c$; the figure shows the median and the interquartile band of $A_r(c)$ over the runs, that is, the 50% empirical attainment surface and its quartiles, following the attainment-based assessment of stochastic multi-objective optimisers [16]. On E1, Green MGGP and MO-lex reach an exact model at the smallest reachable cost in the median run, whereas SO needs a more expensive one, and beyond that cost the surfaces differ only at machine precision. On E2, the surfaces of SO and MO-lex lie marginally below that of Green MGGP in identification at costs between 12 and 14 (median NRMSE 0.0255 against 0.0256), but not in validation, where the Green surface is the lowest at every cost. On E3 the picture is more nuanced: at costs of 18 operations and above, SO and MO-lex attain a lower identification error than Green MGGP, which spreads its population over the whole cost axis instead of concentrating it at the expensive end. This advantage does not carry over to free-run validation, where the Green surface is the lowest at every cost and reaches its smallest median validation error with models of about 20 operations. The baselines thus spend their budget refining expensive models whose additional identification accuracy is not reflected in simulation.

Fig. 6 shows the sets behind these statistics for one representative run of each method, chosen by the rule used throughout, the median validation error of the selected model. The Green sets cover the whole cost range with many models: on E3, nine models between 2 and 18 operations per sample, against six models between 13 and 21 for MO-lex and a single model of 21 operations for SO, and on E2 seven models between 2 and 12, against four for MO-lex. In validation, the representative Green sets also contain the most accurate model of the three runs, with NRMSE 0.0295 at 9 operations on E2 and 0.0617 at 16 operations on E3. On E3 the τ -rule selects instead the Green model of 18 operations, whose validation error is 0.0787, which illustrates on a single run the limitation of a rule based on identification error alone.

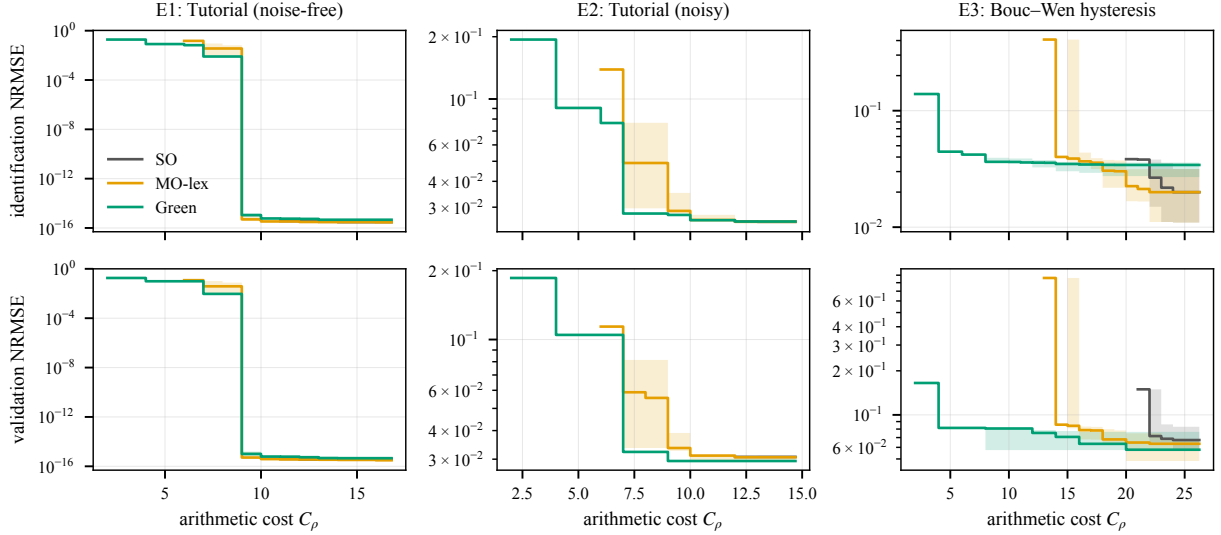


Figure 5: Empirical attainment surfaces of the returned sets with $\rho = 1$: median (lines) and interquartile band (shading) over 30 runs of the smallest identification error (top) and free-run validation error (bottom) achieved within each cost budget. Logarithmic error axes; on E1 the errors are at machine precision once the exact structure is reached.

Table 5: Empirical verification of Theorems 2 and 3 over 30 seeds. Lexicographic reduction: seeds in which SO and MO-lex produce identical populations in all generations; diverging seeds in which the divergence is preceded by an error tie between individuals of different cost, and in which the populations involved contain at least two individuals with $J_E = \infty$; median generation of the first divergence. Non-deterioration: largest $|U_t^1|$ over all runs and generations, bound $L(G, h)$ of (6) and population size μ ; runs with at least one decrease of the population-front hypervolume.

Ex.	Lexicographic reduction				Capacity			Runs with HV decrease		
	identical	tie	$J_E = \infty$	first div.	$\max U_t^1 $	$L(G, h)$	μ	SO	MO-lex	Green
E1	1/30	29/29	29/29	2	9	50	100	30	30	0
E2	1/30	29/29	29/29	2	9	50	100	30	30	0
E3	22/30	8/8	8/8	11	13	260	100	30	30	0

3.2 Search dynamics and empirical verification of the theory

Fig. 7 follows the population of each run over the generations. Its top row shows the hypervolume of the non-dominated subset of the population, with the cost of SO individuals computed *a posteriori*. For Green MGGP this quantity never decreases, as Theorem 3 predicts, and it rises in steps as cheaper or more accurate structures are found. For SO and MO-lex it peaks within the first few generations and then decreases or oscillates, because elitism by error discards cheap non-dominated individuals once more accurate ones appear. The two baseline curves are almost indistinguishable, as anticipated by Theorem 2. The bottom row shows the number of distinct structures in the population: with structural duplicate removal, Green MGGP reaches $\mu = 100$ distinct structures within two generations and keeps them, whereas the median baseline populations hold between about 30 and 80, the rest being copies.

Table 5 tests both theorems directly. SO and MO-lex were run with identical seeds, and their populations were compared generation by generation. On E3, the two runs coincide in all generations for 22 of the 30 seeds, and in each of these seeds MO-lex returns exactly the non-dominated subset of the final SO population. In every one of the remaining seeds, and in

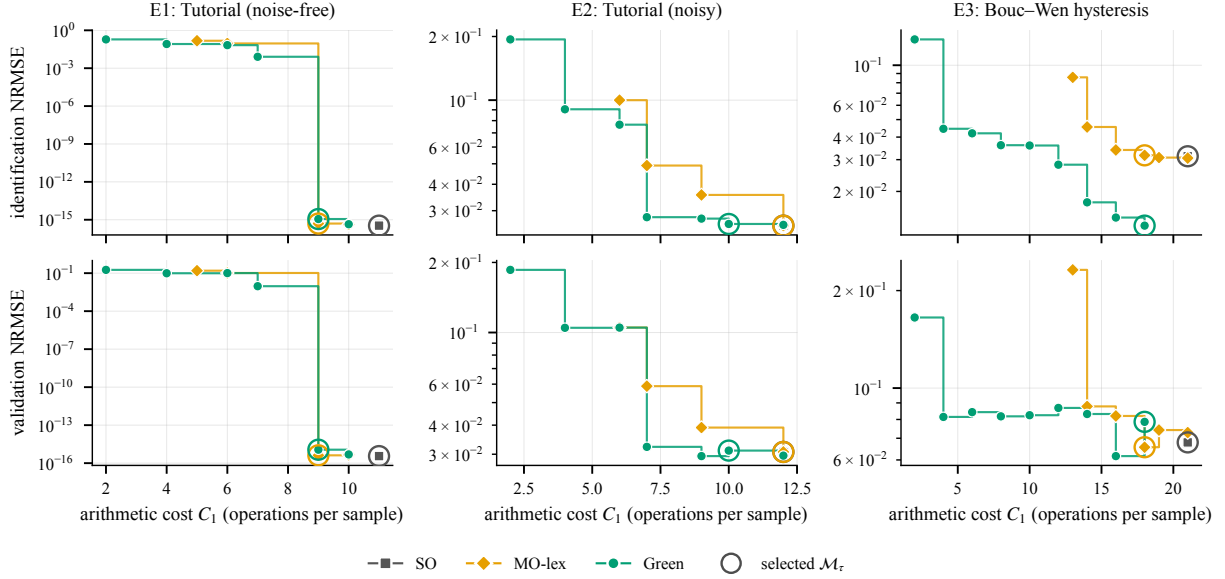


Figure 6: Returned sets of the representative runs with $\rho = 1$ (median validation error of the selected model; seeds 2, 27 and 15 on E1, 3, 26 and 2 on E2, and 22, 27 and 24 on E3, for SO, MO-lex and Green MGGP), plotted against the identification error (top) and the free-run validation error of the same models (bottom). Rings mark the model selected by the τ -rule; errors below 10^{-16} are shown at 10^{-16} .

all diverging seeds of E1 and E2, the divergence is preceded by a tie in J_E between individuals of different cost, which is precisely the case excluded by the hypothesis of Theorem 2, and the populations involved contain at least two individuals with $J_E = \infty$. On E1 and E2, the first divergence occurs at a median of the second generation, yet the baseline curves of Fig. 7 remain close, which suggests that these ties have little effect on the population fronts. For Green MGGP, the first front of the de-duplicated union never exceeded 13 individuals, far below $\mu = 100$, so the hypothesis of Theorem 3 held in every generation of every run; for E1 and E2 it is also guaranteed a priori by $L(G, h) = 50 \leq \mu$. Accordingly, no Green run showed a decrease of the population-front hypervolume, whereas every SO and MO-lex run showed at least one.

3.3 Selected models

When a single model is chosen by the τ -rule, Green MGGP yields a median cost of 9, 10 and 14 operations per sample on E1–E3, against 11, 12 and 20.5 for SO; the ratios of these medians correspond to reductions of 18%, 17% and 32%. The median selected cost of MO-lex equals that of Green MGGP on E1 and E2, but its interquartile range extends to more expensive models on E2, and on E3 its median is the same as that of SO. On E1 and E2 the validation errors of the selected models are practically equal: on E1 all methods reach machine precision, and the statistically detectable differences there concern errors below 10^{-14} ; on E2 the median NRMSE of Green MGGP is 0.031, as for both baselines, and no difference is significant. On E3, the cheaper models selected by Green MGGP have a higher median validation error (0.078 against 0.067 for SO and 0.066 for MO-lex), although the difference is not significant (Holm-adjusted $p > 0.2$); the rule selects within 5% of the best identification error of each set, and the Green sets on E3 do not contain the most accurate expensive models found by the baselines.

The representative models illustrate individual runs rather than the reductions between medians. On E3, the Green and MO-lex representatives have equal cost, and Green has higher validation NRMSE (Table 6); this example does not show a cost advantage over MO-lex. On E2, the Green representative includes the spurious squared delayed-output term. The E2 panel of

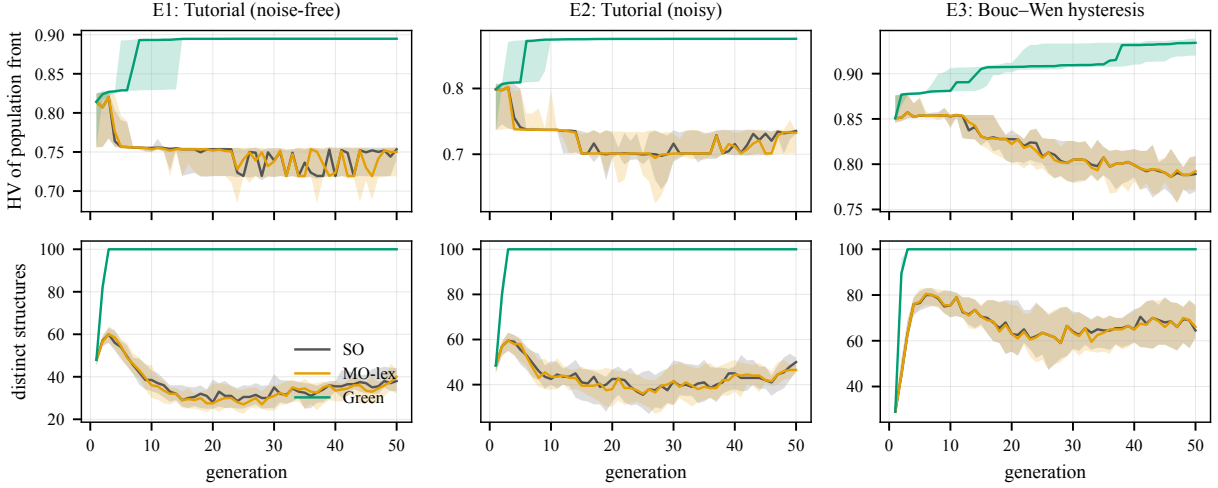


Figure 7: Search dynamics with $\rho = 1$: median (lines) and interquartile band (shading) over 30 runs of the hypervolume of the population front (top; for SO the cost is computed a posteriori) and of the number of distinct structures in the population (bottom).

Table 6: Regressors of the model selected by the τ -rule in the representative run (median validation NRMSE of the selected model) and its free-run validation NRMSE.

Ex.	Method	Regressors	C	NRMSE
E2	SO	$u(k), u(k)^2, u(k-2), y(k-1), y(k-1)y(k-2)$	12	0.031
E2	MO-lex	$u(k), u(k)^2, u(k-2), y(k-1), y(k-1)y(k-2)$	12	0.031
E2	Green	$u(k), u(k)u(k-2), y(k-1), y(k-2)^2$	10	0.031
E3	SO	$u(k), u(k)^3, u(k-3)^2, u(k-6), u(k-7), y(k-1)^3, y(k-2), y(k-7)$	21	0.068
E3	MO-lex	$u(k), u(k)^3, u(k-3), u(k-5), u(k-8), y(k-1)^3, y(k-6)$	18	0.066
E3	Green	$u(k), u(k)u(k-2)^2, u(k)^2u(k-5), u(k-5), u(k-6), y(k-1), y(k-6)$	18	0.079

Fig. 9 shows the similar validation trajectories despite these different regressors; visual agreement does not establish structure recovery.

Structure recovery shows both the strength of the trade-off set and the limits of a single selection rule. On E1, all methods select a model that contains the three true regressors in every run, with one additional term for Green MGGP and MO-lex and one or two for SO. The additional term is imposed by the package, which cannot evaluate the undelayed representation containing output factors without a backshift operator, so the exact three-term structure ($C = 7$) is not reachable and 9 is the smallest exact model. On E2, the noise makes models with spurious terms reduce J_E by slightly more than the 5% tolerance, and no method selects a model containing the true regressors. However, the returned set of Green MGGP contains such a model in all runs ($\text{Rec.}p = 1.00$), against none of the SO runs and 13% of the MO-lex runs. The trade-off set therefore retains the physically meaningful structure, and its absence from $\mathcal{M}_\tau$ is a property of the rule, which uses only the identification error, rather than of the search. On E3 (Table 6, Fig. 9), the selected models reproduce the hysteresis loop, and the selected structures combine delayed outputs and inputs in products of degree up to three.

Fig. 8 resolves these statements by regressor. On E2, $u(k)$ and $y(k-1)$ enter the selected model of every method in every run, but the bilinear term $u(k)y(k-1)$ never does; it is present in every Green set, in 13% of the MO-lex sets and in no SO run. In its place, the selected Green models use $u(k)u(k-2)$ in 93% of the runs, whereas SO keeps the quadratic term $u(k)^2$ in 93% of the runs. On E3, where no true structure is known, the input terms $u(k)$, $u(k-8)$ and $u(k)^3$

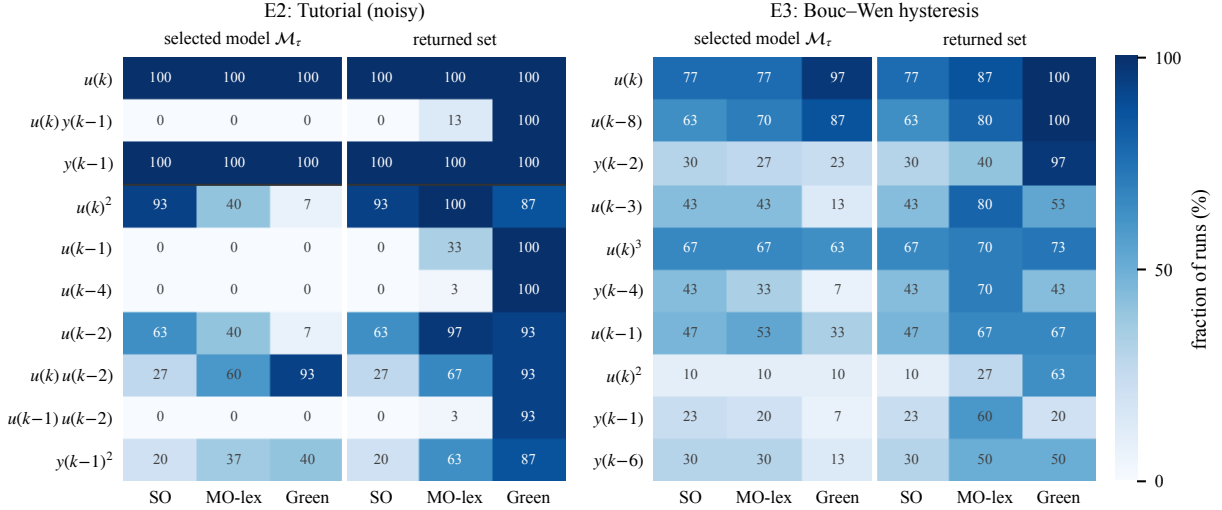


Figure 8: Percentage of the 30 runs with $\rho = 1$ in which each regressor appears in the selected model $\mathcal{M}_\tau$ (left block of each panel) and in at least one model of the returned set (right block), for the ten most frequent regressors. On E2, the three regressors above the line form the true structure (10).

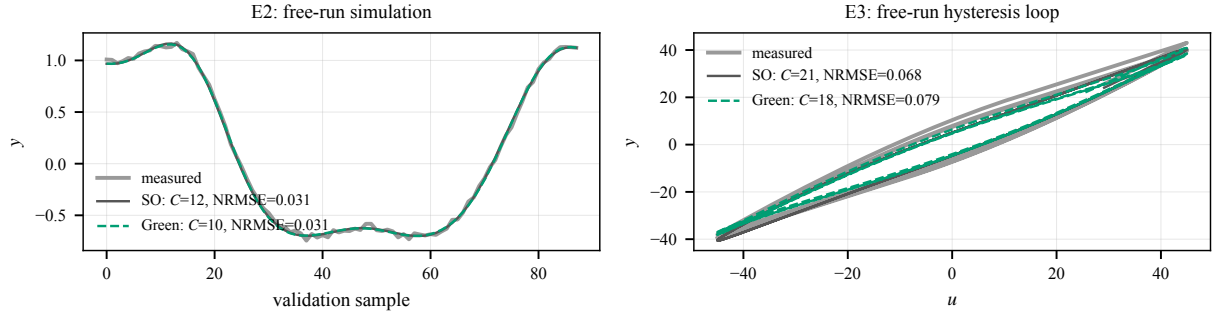


Figure 9: Free-run validation of the selected models of the representative runs with $\rho = 1$: E2 time series (left) and E3 input–output loop (right).

recur in the selected models of all methods, and the Green sets contain delayed outputs more often than the baselines, for instance $y(k-2)$ in 97% of the runs against at most 40%, so that the search retains structurally different alternatives that the selection rule does not reach.

3.4 Sensitivity to the selection tolerance

The tolerance $\tau = 0.05$ was fixed before the experiments; Fig. 10 examines how the selected models change when it varies between 0 and 1, without altering the reported results. In agreement with Proposition 3(ii), the selected cost of the Pareto-based configurations decreases monotonically with τ , whereas SO is unaffected because it returns a single model. The range of costs that Green MGGP offers is much wider: on E3, its median selected cost falls from 14 operations at $\tau = 0.05$ to 4 at $\tau \geq 0.5$, while that of MO-lex stays at or above 16. The price is a gradual increase in validation error, from 0.078 to 0.082 on E3 and from 0.031 to 0.032 on E2. At $\tau = 0.1$ on E2, the median selected cost equals 7, but equality of cost does not imply equality of structure. Models at this cost omit the true bilinear regressor; the 30% of Green runs that recover all true regressors select more expensive models. Recovery is therefore not monotone in τ . The dotted lines show the smallest validation error available in the Green sets: on E2 and E3 it is below the median validation error of the selected model of every configuration, which confirms that the sets contain better models than any rule based on identification data alone

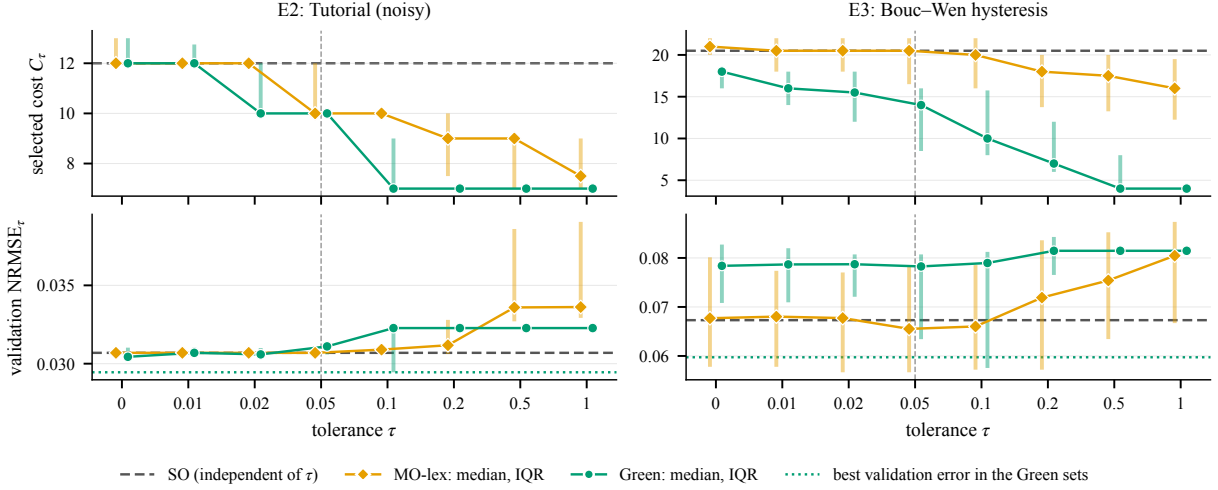


Figure 10: Sensitivity of the τ -rule with $\rho = 1$ on E2 and E3 (on E1 the selected model does not depend on τ): median (markers) and interquartile range (bars) over 30 runs of the selected cost (top) and of its free-run validation NRMSE (bottom) for MO-lex and Green MGGP. The dashed horizontal line is SO, which returns a single model; the vertical dashed line marks the declared value $\tau = 0.05$, and the dotted line is the median of the smallest validation error in the Green sets.

can identify.

3.5 Effect of the multiplication weight

All configurations were also run with the bit-level weight ρ_K , which reproduces the Green-Box complexity up to the factor B (Proposition 2); those runs reproduced bit for bit an earlier set of runs with the complexity $Ba + B^\alpha b$, and the SO runs, which do not use the cost, were identical under both weights. Table 7 summarises the comparison. The conclusions on the selection scheme do not depend on the weight: under ρ_K , Green MGGP again exceeds both baselines in identification and validation hypervolume in every run ($A_{12} = 1.00$), with medians of 0.929, 0.908 and 0.946 against at most 0.830, 0.811 and 0.848, and its selected models are 27%, 15% and 23% cheaper than those of SO in Karatsuba-weighted cost, against 18%, 17% and 32% in operations under $\rho = 1$.

Fig. 11 locates the selected models in the plane of additions and multiplications. Lines of constant C_1 have slope -1 , whereas lines of constant C_{ρ_K} are almost horizontal, so that under ρ_K a structure is cheaper essentially when it has fewer multiplications, whatever its number of terms, while under $\rho = 1$ terms and degrees are exchanged one for one. On E1 and E2 the Green selections under both weights concentrate on the same point, $(a, b) = (4, 5)$ and $(4, 6)$, below and to the left of all SO selections. On E3 they spread over the lattice, from three to eight terms, and the most frequent selection differs between the weights, $(5, 9)$ for $\rho = 1$ and $(6, 10)$ for ρ_K , whereas the SO models use seven or eight terms and between 10 and 17 multiplications.

The weight does change which structures are selected. On E1 the same structure is selected under both weights in every run, but on E2 this holds in 60% of the runs and on E3 in only 3%. With $\rho = 1$, adding a linear term costs as much as raising the degree of an existing one, whereas with ρ_K a multiplication costs about eleven additions, so the two weights favour different shapes of polynomial. On E3 the models selected with $\rho = 1$ need 14 operations per sample against 16 for those selected with ρ_K , and they are also cheaper when costed with ρ_K (102.3 against 119.9), at a higher median validation error (0.078 against 0.071). Because the identified structures depend on ρ , the weight is a property of the target platform that should be set before identification; Section 4 shows that, on the platform used here, measured execution

Table 7: Effect of the multiplication weight ρ over 30 runs (medians). HV: identification hyper-volume; ΔC_τ : reduction of the median selected cost of Green MGGP relative to SO, with the cost of the same scenario; $a+b$ and C_{ρ_K} : operation count and Karatsuba-weighted cost of the selected Green model; same: fraction of seeds in which Green MGGP selects the same structure under both weights.

Ex.	weight	HV _{id}			Green $\mathcal{M}_\tau$			
		SO	MO-lex	Green	ΔC_τ	$a+b$	C_{ρ_K}	same
E1	$\rho = 1$	0.560	0.750	0.895	18%	9	61.0	100%
	ρ_K	0.640	0.830	0.929	27%	9	61.0	
E2	$\rho = 1$	0.507	0.733	0.876	17%	10	72.3	60%
	ρ_K	0.620	0.811	0.908	15%	10	72.3	
E3	$\rho = 1$	0.698	0.792	0.934	32%	14	102.3	3%
	ρ_K	0.769	0.848	0.946	23%	16	119.9	

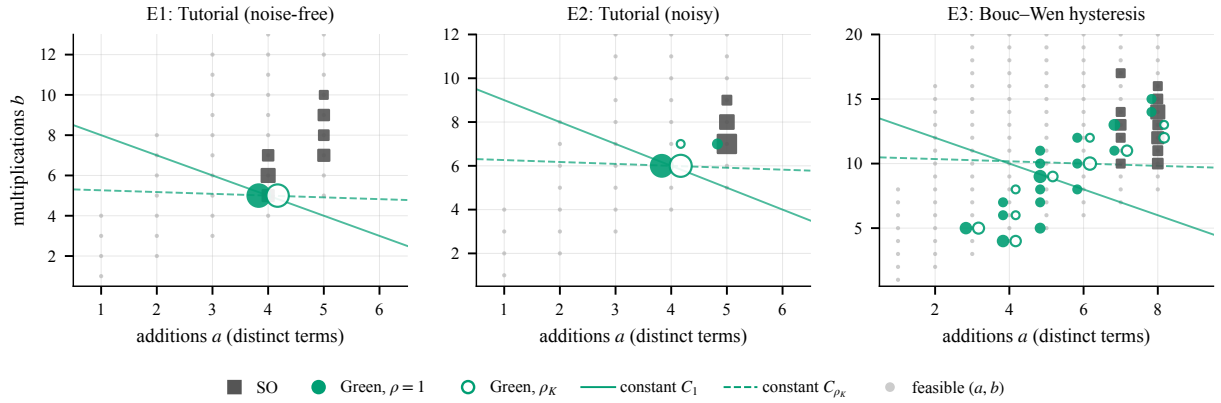


Figure 11: Selected models in the plane of additions a and multiplications b over 30 runs: SO, and Green MGGP with $\rho = 1$ and with ρ_K (marker area proportional to the number of runs; the two Green weights are offset horizontally for visibility). Grey dots are the feasible pairs counted by $L(G, h)$ in Proposition 1; the lines pass through the most frequent Green selection with constant C_1 (solid) and constant C_{ρ_K} (dashed).

supports $\rho = 1$ rather than ρ_K .

3.6 Computational effort and limitations

The median run times of Green MGGP were 2.8, 2.9 and 5.2 s, comparable to those of SO (2.0, 2.2 and 5.3 s) and lower than those of MO-lex (3.4, 3.8 and 6.9 s); the extra sorting is negligible compared with model evaluation. These are software timings of the identification procedure; the time and energy needed to execute the identified models are examined separately in Section 4. The study is also bounded by the package itself. Besides the backshift requirement above, its simulation routines fail for products of delayed outputs and inputs whenever the maximum delay exceeds one, so such individuals receive infinite error in all three configurations; these failures are also the source of the ties that break the hypothesis of Theorem 2. A further implementation error affects output factors at the largest delay: the circular output buffer can substitute $y(k-1)$ for the intended delayed output, as detailed in Section 1. Consequently, some stored simulation errors are not errors of the intended polynomial. The selected models are numerically unaffected within the reported simulator agreement, but this does not establish that the returned sets or search trajectories are unaffected. All methods use the same predictor;

correcting it would require repeating the searches. The present results therefore characterise this implementation and its restrictions. The generation counts are equal, but Green produces more offspring per generation; equal-evaluation comparisons remain future work. Pareto trade-offs between accuracy and complexity are already central to multigene symbolic regression in GPTIPS [2], and complexity objectives in symbolic regression extend to the order of nonlinearity [17]. The present comparison concerns the operation cost of the duplicate-free polynomial and its treatment by the selection scheme. An ablation with Pareto selection and term or node count is still needed to isolate the contribution of the arithmetic objective from the selection scheme. Finally, the examples are small, and the classification mode, which returns a single objective, was not considered.

4 Execution Time and Energy of the Selected Models

The cost C_ρ in (5) counts the operations of the identified polynomial with a weight ρ that must reflect the platform; it does not by itself establish that cheaper structures run faster or consume less energy. This section reports historical measurements of free-run simulation over the validation data. Timing characterises the relation between operations and runtime; power integration estimates energy; carbon intensity converts energy to emissions. These measurements read stored models and do not alter the search objectives or its results. The corrected models require a new session, which has not yet been performed.

4.1 Reconstruction and execution implementations

For every run of SO, and of MO-lex and Green MGGP under both weights, the model $\mathcal{M}_\tau$ was rebuilt, re-estimated on the identification data and simulated in free run over the validation input, which is the computation behind the validation NRMSE of Table 3. The rebuilt models reproduced the stored cost and validation error exactly for 449 of the 450 runs; the remaining SO run on E3, whose validation simulation had failed, is treated as missing. Each model was executed by two implementations whose outputs agreed to within 6×10^{-13} . The *canonical* simulator is straight-line Python code generated from the duplicate-free polynomial, so that each sample executes exactly the $a(\mathcal{R})$ additions and $b(\mathcal{R})$ multiplications counted by C_ρ . The *package* simulator is the free-run predictor of the `mggp` package, which evaluates every gene, duplicates included, on windows of NumPy arrays; it represents the cost incurred by a user of the package as distributed.¹

4.2 Timing and power measurement

The execution time of each model was taken as the median of five batches of repeated simulations, each lasting at least 50 ms. Energy was measured on an Apple M2 system-on-chip with Python 3.13 on mains power, from the combined CPU, GPU and neural-engine power reported by `powermetrics` every 100 ms. One measurement block repeated complete cycles over the selected models of one example, configuration and simulator, so that every run carried the same weight, for at least 10 s. Each of 10 rounds visited the 30 groups in a seeded random order and was bracketed by idle blocks of the same length, giving 311 blocks in about 52 min. The energy of one simulation is

$$E_{\text{sim}} = \frac{E_{\text{block}} - \tilde{P}_{\text{idle}} t_{\text{block}}}{n_{\text{sim}}}, \quad (11)$$

where $\tilde{P}_{\text{idle}}$ is the median power of all idle blocks, chosen before the measurement because it is robust to an idle block disturbed by background activity. The idle power ranged from 0.2

¹The package keeps only n past outputs in its free-run window and delays them circularly, so a factor $y(k-1-n)$ is read as $y(k-1)$; the canonical simulator uses the intended delays. Among the selected models this affects only genes whose coefficients are below 10^{-15} , hence the agreement of the two outputs.

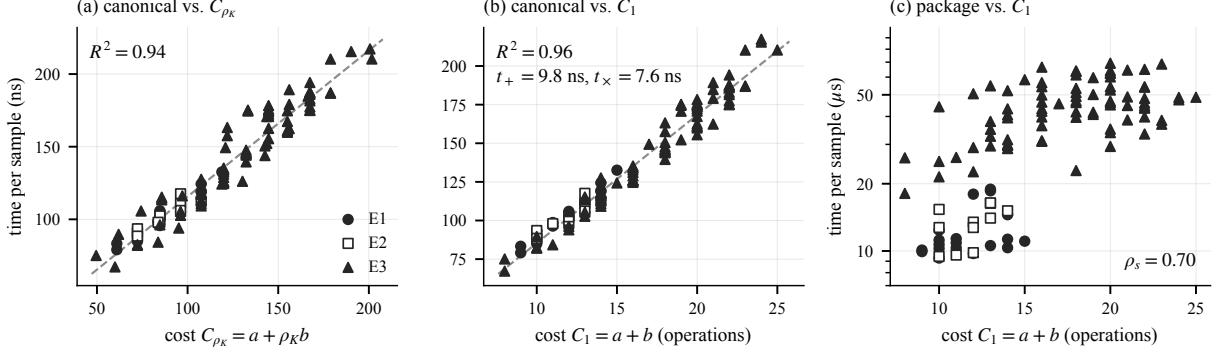


Figure 12: Execution time per simulated sample of the 114 distinct canonical polynomials selected over all runs and both weights, against the Karatsuba-weighted cost C_{ρ_K} (a) and the operation count $C_1 = a + b$ (b), with least-squares lines; (c) the same models executed by the package free-run predictor (logarithmic scale).

to 0.4 W, and subtracting instead the mean of the two idle blocks around each round changes the energy ratios reported below by less than 0.05 percentage points. The energy equation uses measured power and elapsed time, not the arithmetic objective. Carbon conversion is $\text{CO}_2\text{e} = E_{\text{sim}} I_{\text{grid}}$, with energy expressed in kWh. Emissions follow from E_{sim} and the grid carbon intensity of Ireland, 291 gCO₂e/kWh, taken from the CodeCarbon energy-mix data [18]. CodeCarbon supplies the carbon-intensity factor only; energy is measured by **powermetrics**. Green MGGP with $\rho = 1$ is compared with SO, with MO-lex ($\rho = 1$) and with Green MGGP under ρ_K over the rounds, with the tests of Section 2, and energy ratios to SO are formed within each round.

4.3 Time-based assessment of the declared weights

Fig. 12 relates the execution time per simulated sample to the structure of the 114 distinct canonical polynomials selected in all runs. For the canonical simulator, a least-squares fit of $t = t_0 + t_+ a + t_\times b$ explains the times with $R^2 = 0.97$, with $t_+ = 9.8$ ns per addition and $t_\times = 7.6$ ns per multiplication. These fitted coefficients include CPython execution and dispatch overhead and come from search-selected structures, rather than an independent arithmetic microbenchmark. They characterise this implementation on this laptop, not instruction-level hardware costs. The measured weight $t_\times/t_+ = 0.78$ is of order one, as was the value of 0.98 obtained in a pilot session with the models selected under ρ_K , and an order of magnitude below $\rho_K \approx 11.39$. Accordingly, the operation count C_1 predicts the time better ($R^2 = 0.96$) than the bit-level cost C_{ρ_K} ($R^2 = 0.94$): on this interpreted binary64 implementation, the execution is dominated by the dispatch of each operation rather than by the bit-level work of multiplication. The unit weight is used rather than the fitted ratio because the latter reflects implementation details of the interpreter, such as the indexing of delayed values, that differ between sessions and platforms.

The package simulator is about 169 times slower and its time is only weakly related to the cost (Spearman correlation 0.70 with C_1 , $R^2 = 0.50$). It evaluates every gene of the individual, including duplicates and genes with negligible coefficients, through array operations whose overhead does not depend on the degree, so its time is governed by the number and size of the gene trees rather than by the executed structure.

4.4 Energy and emissions

The net power drawn during simulation was nearly constant, about 6.7 W for the canonical simulator and between 6.2 and 6.5 W for the package simulator, so the energy of a simulation follows its execution time. Table 8 and Fig. 13 summarise the measurements. With the canonical

simulator, the models selected by Green MGGP required less energy than the SO models on every example, with median per-round ratios of 0.81, 0.87 and 0.65 and complete separation of the rounds ($A_{12} = 0.00$). They also required less energy than the MO-lex models on E2 and E3, whereas on E1 both methods select the same nine-operation structures and their energies do not differ significantly. The reductions follow the operation counts of Table 3 closely: on E3, for instance, the median counts of 14 and 20.5 operations per sample are in the ratio 0.68, and the measured energy ratio is 0.65.

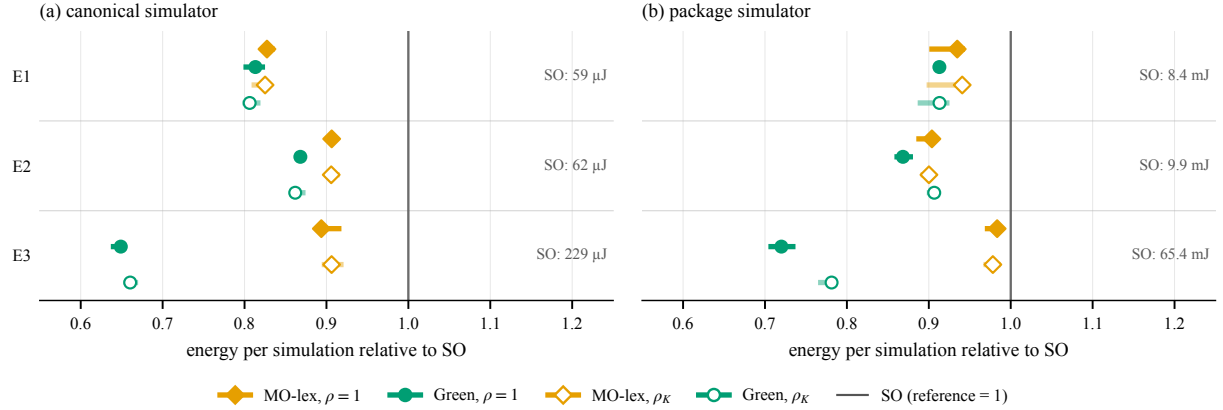


Figure 13: Energy per free-run simulation of the selected models relative to SO, formed within each of the 10 measurement rounds: median (markers) and interquartile range (bars) over the rounds, for the canonical (a) and package (b) simulators. Filled markers: $\rho = 1$; open markers: ρ_K . The absolute SO energies are given on the right.

Table 8: Free-run simulation of the selected models on the validation data: median over 10 measurement rounds of the time and idle-subtracted energy per simulation, and the corresponding emissions per 10^6 simulations at 291 gCO₂e/kWh. MO-lex and Green use $\rho = 1$ unless marked ρ_K ; $a+b$: median number of operations per sample of the selected models. *, †, ‡: Green MGGP ($\rho = 1$) differs from SO, MO-lex, Green (ρ_K) (Holm-adjusted $p < 0.05$).

Ex.	Method	$a+b$	canonical simulator			package simulator		
			t (μ s)	E (μ J)	CO ₂ e (mg)	t (ms)	E (mJ)	CO ₂ e (g)
E1	SO	11	8.9	59	4.8	1.35	8.4	0.67
	MO-lex	9	7.3	49	4.0	1.24	7.8	0.63
	Green	9	7.2	48*	3.9	1.21	7.6*	0.62
	MO-lex (ρ_K)	9	7.3	49	3.9	1.25	7.9	0.63
	Green (ρ_K)	9	7.2	48	3.9	1.22	7.6	0.62
E2	SO	12	9.3	62	5.0	1.60	9.9	0.80
	MO-lex	10	8.4	56	4.5	1.42	8.9	0.72
	Green	10	8.0	54*†	4.3	1.37	8.6*†‡	0.70
	MO-lex (ρ_K)	10	8.4	56	4.5	1.42	8.9	0.72
	Green (ρ_K)	10	8.0	53	4.3	1.43	9.0	0.73
E3	SO	20.5	33.7	229	18.5	10.21	65.4	5.28
	MO-lex	20.5	30.8	205	16.5	9.99	63.9	5.16
	Green	14	21.9	147*†‡	11.9	7.25	47.0*†‡	3.80
	MO-lex (ρ_K)	20.5	30.8	205	16.6	9.99	64.2	5.18
	Green (ρ_K)	16	22.5	150	12.1	7.91	51.0	4.12

The weight used during identification matters where it changes the selected structures. On E1 and E2, where the two weights mostly select the same or equally long models, Green MGGP

with $\rho = 1$ and with ρ_K consume the same energy with the canonical simulator. On E3, the models selected with $\rho = 1$ need 2% less energy than those selected with ρ_K with the canonical simulator and 8% less with the package simulator, both significant, so the unit-weight scenario yields structures that are also cheaper to execute in this historical session.

With the package simulator, the advantage over SO is smaller but significant on every example, with ratios of 0.91, 0.87 and 0.72, and over MO-lex on E2 and E3. Expressed per 10^6 simulations of the validation record, the E3 models selected by Green MGGP correspond to 11.9 mg CO₂e with the canonical simulator, against 18.5 mg for SO, and to 3.80 g against 5.28 g with the package simulator. The implementation thus matters more than the structure: the same models differ by two orders of magnitude between the two simulators, and the benefit of a cheaper structure is realised in full only when the canonical polynomial is executed.

These measurements concern interpreted code on one laptop system-on-chip in one session, not the embedded processors on which identified models are often deployed and where multiplications may cost more. They show that the ordering induced by C_ρ carries over to measured time and energy for canonical execution, and that ρ should be calibrated on the target platform before identification.

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